PRE-ALGEBRA (7TH) · UNIT 7: SOLVING EQUATIONS AND INEQUALITIES

Summer Job Savings

Compare summer jobs and savings plans with multi-step equations, inequalities, and interest.

CP2

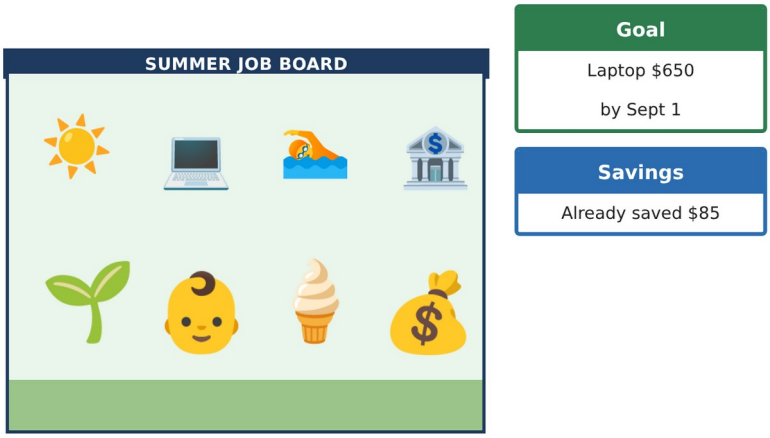
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You want to buy a $650 laptop by the end of the summer. You can mow lawns, babysit, or work at the snack stand at the town pool, and you can put your money in a savings account that earns interest.

You will write and solve equations and inequalities to compare jobs, reach your goal, and plan your savings.



Learning Goals

- I can write and solve two-step equations.
- I can solve two-step inequalities.
- I can use the simple interest formula.

Project Timeline

When	What you will do
Day 1	Earn the Money
Day 2	Reach the Goal
Day 3	Save with Interest
Final	Summer Savings Plan + reflection

Part 1: Earn the Money

Day 1 · Use the Job Offer Cards. You have $85 saved.

Job Offer Cards

Lawn Mowing	Babysitting	Snack Stand
$15 per lawn	$12 per hour	$10.50 per hour
$40 for a mower tune-up	$20 CPR class first	+ $25 signing bonus

Dog Walking	Bank A	Bank B
$8 per walk	3% simple interest	2.8% compounded
no start-up cost	per year	yearly

1. Mowing: write an expression for the money earned from m lawns after paying for the $40 tune-up.

   Starter: *15 × ___ − 40*

2. How many lawns must you mow to earn $200 after the tune-up? Solve 15m − 40 = 200.

3. Babysitting: solve 12h − 20 = 172 to find the hours needed.

4. Check your mowing answer by substituting.

Part 2: Reach the Goal

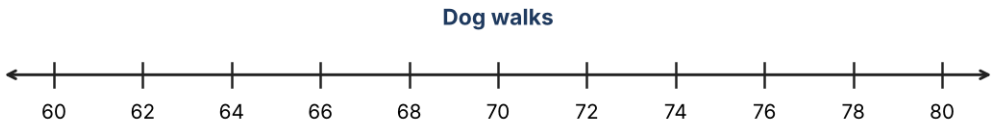
Day 2 · You need $650 and have $85.

5. Write and solve an inequality for the number of dog walks w you need: 85 + 8w ≥ 650.

6. Snack stand: 85 + 25 + 10.50h ≥ 650. Solve for h.

7. If you work 12 hours a week at the snack stand, how many weeks is that?

8. Graph the solution of w ≥ 71 on a number line.



Part 3: Save with Interest

Day 3 · Simple interest: I = Prt.

9. You put $300 in Bank A at 3% for 1 year. How much interest do you earn?

10. How much is in the account after 2 years with simple interest?

11. Choose a job and write your full summer plan with an equation or inequality.

Final Product: Summary Savings Plan

Your final product must include:

- Your job choice and earnings equation
- An inequality for reaching $650
- A graph of the solution
- Interest calculation
- Savings calendar for the summer

Reflection

R1. Why do you undo addition or subtraction first when solving a two-step equation?

R2. Why do you round 70.6 walks up to 71?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Equations and inequalities model each job and goal correctly and support a realistic savings plan. Goes beyond the minimum with a creative, realistic choice.	Equations and inequalities model each job and goal correctly and support a realistic savings plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes the job choice, solved equations and inequalities, interest comparison, and a calendar. Polished, organized, and visually engaging.	Plan includes the job choice, solved equations and inequalities, interest comparison, and a calendar. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.